

Explore signatures and logs with Suricata

Activity 1

Explore custom rules

```
analyst@1a30f9201cb8:~$ cat custom.rules
alert http $HOME_NET any -> $EXTERNAL_NET any (msg:"GET on wire"; flow:established,to_server; content:"GET"; http_method; sid:12345; rev:3;)
analyst@1a30f9201cb8:~$
```

Signature contains three components namely action, header and rule options

Component 1

Action

```
alert http $HOME_NET any -> $EXTERNAL_NET  
any (msg:"GET on wire";  
flow:established,to_server; content:"GET";  
http_method; sid:12345; rev:3;)
```

Component 2

Header

```
alert http $HOME_NET any -> $EXTERNAL_NET  
any (msg:"GET on wire";  
flow:established,to_server; content:"GET";  
http_method; sid:12345; rev:3;)
```

Component 3

Rule Options

```
alert http $HOME_NET any -> $EXTERNAL_NET  
any (msg:"GET on wire";  
flow:established,to_server; content:"GET";  
http_method; sid:12345; rev:3;)
```

Activity 2

Run Suricata with custom rule, trigger it and examine output logs with **fast.log** file

```
analyst@3ce2388195e3:~$ ls -l
total 64
-rw-r--r-- 1 analyst root 141 Feb 12 08:46 custom.rules
-rw-r--r-- 1 root root 57462 Feb 12 08:46 sample.pcap
analyst@3ce2388195e3:~$ ls -l /var/log/suricata
total 16
-rw-r--r-- 1 root root 1419 Feb 12 09:27 eve.json
-rw-r--r-- 1 root root 292 Feb 12 09:27 fast.log
-rw-r--r-- 1 root root 3241 Feb 12 09:27 stats.log
-rw-r--r-- 1 root root 1512 Feb 12 09:27 suricata.log
analyst@3ce2388195e3:~$ cat /var/log/suricata/fast.log
11/23/2022-12:38:34.624866  [**] [1:12345:3] GET on wire [**] [Classification:
 (null)] [Priority: 3] {TCP} 172.21.224.2:49652 -> 142.250.1.139:80
11/23/2022-12:38:58.958203  [**] [1:12345:3] GET on wire [**] [Classification:
 (null)] [Priority: 3] {TCP} 172.21.224.2:58494 -> 142.250.1.102:80
analyst@3ce2388195e3:~$
```

Activity 3

Examine output of **eve.json** log file

Using the jq command, entries are displayed in improved format

```
analyst@3ce2388195e3:/var/log/suricata$ ls -l /var/log/suricata/eve.json
-rw-r--r-- 1 root root 1419 Feb 12 09:27 /var/log/suricata/eve.json
analyst@3ce2388195e3:/var/log/suricata$ jq . /var/log/suricata/eve.json | less
{
  "timestamp": "2022-11-23T12:38:34.624866+0000",
  "flow_id": 1043941533841557,
  "pcap_cnt": 70,
  "event_type": "alert",
  "src_ip": "172.21.224.2",
  "src_port": 49652,
  "dest_ip": "142.250.1.139",
  "dest_port": 80,
  "proto": "TCP",
  "tx_id": 0,
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 12345,
    "rev": 3,
    "signature": "GET on wire",
    "category": "",
    "severity": 3
  },
  "http": {
    "hostname": "opensource.google.com",
    "url": "/",
    "http_user_agent": "curl/7.74.0",
    "http_content_type": "text/html",
    "http_method": "GET",
    "protocol": "HTTP/1.1",
  }
}
```

```
"length": 223
},
"app_proto": "http",
"flow": {
  "pkts_toserver": 4,
  "pkts_toclient": 3,
  "bytes_toserver": 357,
  "bytes_toclient": 788,
  "start": "2022-11-23T12:38:34.620693+0000"
}
```

```
"timestamp": "2022-11-23T12:38:58.958203+0000",
"flow_id": 1644517549839604,
"pcap_cnt": 151,
"event_type": "alert",
"src_ip": "172.21.224.2",
"src_port": 58494,
"dest_ip": "142.250.1.102",
"dest_port": 80,
"proto": "TCP",
"tx_id": 0,
"alert": {
  "action": "allowed",
  "gid": 1,
  "signature_id": 12345,
  "rev": 3,
  "signature": "GET on wire",
  "category": "",

```

```

"alert": {
  "action": "allowed",
  "gid": 1,
  "signature_id": 12345,
  "rev": 3,
  "signature": "GET on wire",
  "category": "",
  "severity": 3
},
"http": {
  "hostname": "opensource.google.com",
  "url": "/",
  "http_user_agent": "curl/7.74.0",
  "http_content_type": "text/html",
  "http_method": "GET",
  "protocol": "HTTP/1.1",
  "status": 301,
  "redirect": "https://opensource.google/",
  "length": 223
},
"app_proto": "http",
"flow": {
  "pkts_toserver": 4,
  "pkts_toclient": 3,
  "bytes_toserver": 357,
  "bytes_toclient": 797,
  "start": "2022-11-23T12:38:58.955636+0000"
}
}
(END)

```

Extract specific event data from eve.json file

```

analyst@3ce2388195e3:/var/log/suricata$
analyst@3ce2388195e3:/var/log/suricata$ jq -c "[.timestamp,.flow_id,.alert.signature,.proto,.dest_ip]" /var/log/suricata/eve.json
["2022-11-23T12:38:34.624866+0000",1043941533841557,"GET on wire","TCP","142.250.1.139"]
["2022-11-23T12:38:58.958203+0000",1644517549839604,"GET on wire","TCP","142.250.1.102"]
analyst@3ce2388195e3:/var/log/suricata$

```

Display event logs related to a specific flow_id

```
analyst@3ce2388195e3:/var/log/suricata$ jq "select(.flow_id==1043941533841557)" /var/log/suricata/eve.json
{
  "timestamp": "2022-11-23T12:38:34.624866+0000",
  "flow_id": 1043941533841557,
  "pcap_cnt": 70,
  "event_type": "alert",
  "src_ip": "172.21.224.2",
  "src_port": 49652,
  "dest_ip": "142.250.1.139",
  "dest_port": 80,
  "proto": "TCP",
  "tx_id": 0,
  "alert": {
    "action": "allowed",
    "gid": 1,
    "signature_id": 12345,
    "rev": 3,
    "signature": "GET on wire",
    "category": "",
    "severity": 3
  },
  "http": {
    "hostname": "opensource.google.com",
    "url": "/",
    "http_user_agent": "curl/7.74.0",
```

```
    "http_content_type": "text/html",
    "http_method": "GET",
    "protocol": "HTTP/1.1",
    "status": 301,
    "redirect": "https://opensource.google/",
    "length": 223
  },
  "app_proto": "http",
  "flow": {
    "pkts_toserver": 4,
    "pkts_toclient": 3,
    "bytes_toserver": 357,
    "bytes_toclient": 788,
    "start": "2022-11-23T12:38:34.620693+0000"
  }
}
analyst@3ce2388195e3:/var/log/suricata$
```